

The Most Beautiful Inequality In The Universe

$$(\mathbb{E}[XY])^2 \leq \mathbb{E}[X^2] \mathbb{E}[Y^2]$$

Why correlation cannot exceed geometry

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The Rule



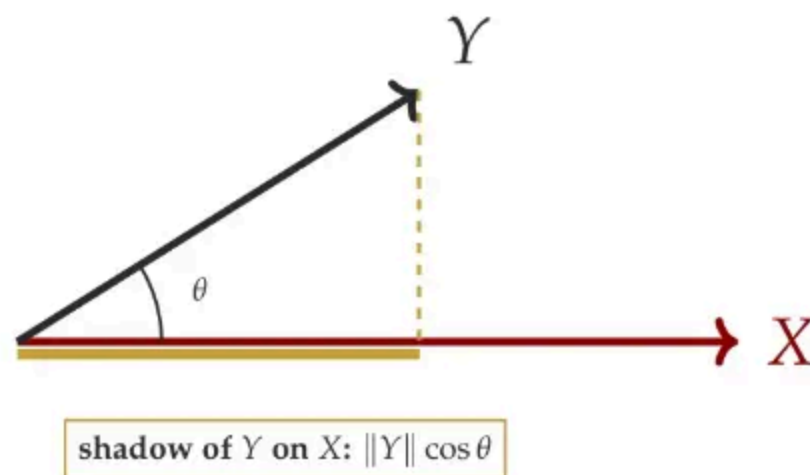
Two random numbers X and Y can move together, but only so much. Their average product $\mathbb{E}[XY]$ is capped by how big each one is on its own.

CAUCHY-SCHWARZ INEQUALITY

If X and Y have finite second moments, then

$$(\mathbb{E}[XY])^2 \leq \mathbb{E}[X^2] \mathbb{E}[Y^2].$$

In geometric words, $|\langle X, Y \rangle| \leq \|X\| \|Y\|$ with $\langle X, Y \rangle = \mathbb{E}[XY]$ and $\|X\| = \sqrt{\mathbb{E}[X^2]}$.



Random variables are vectors: the shadow of one on the other is never longer than the vector itself

$$|\rho(X, Y)| = |\cos \theta| \leq 1$$

Why it is underrated. Every correlation you have ever seen lives between -1 and 1 because of this single line: a cosine cannot leave $[-1, 1]$.

Why It Is True



The proof needs no geometry at all, only the fact that a square is never negative.

STEP 1: SQUARE SOMETHING

For every real number t , $\mathbb{E}[(X + tY)^2] \geq 0$. Expanding,

$$q(t) = t^2 \mathbb{E}[Y^2] + 2t \mathbb{E}[XY] + \mathbb{E}[X^2] \geq 0.$$

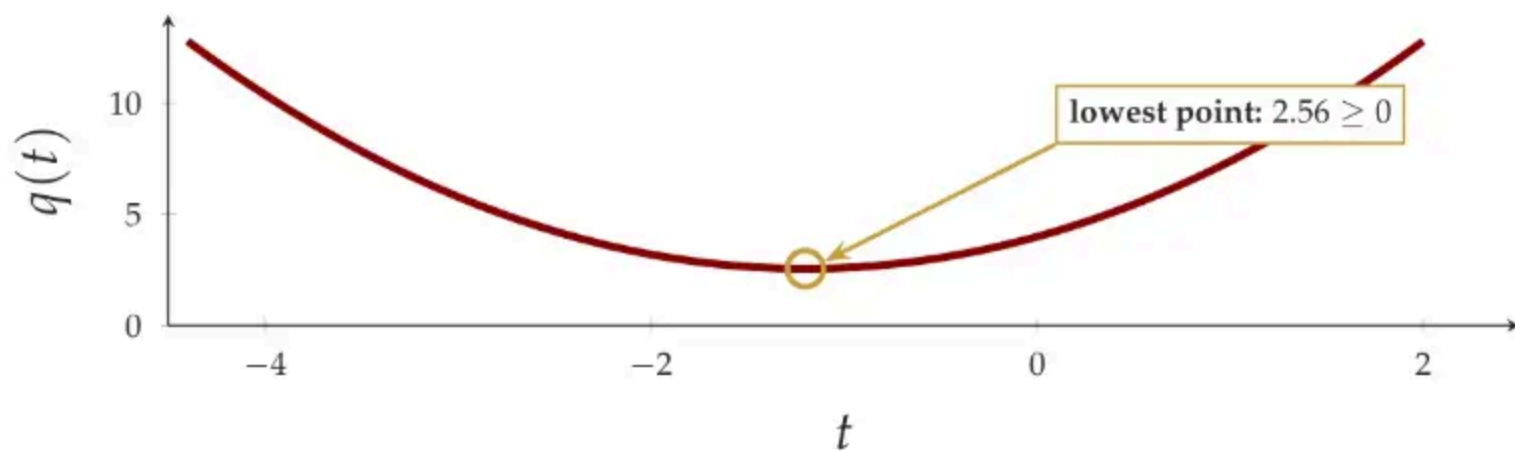
STEP 2: A PARABOLA THAT NEVER GOES BELOW ZERO

A quadratic that is never negative cannot cross the axis, so its discriminant is at most zero:

$$(2 \mathbb{E}[XY])^2 - 4 \mathbb{E}[Y^2] \mathbb{E}[X^2] \leq 0.$$

STEP 3: DIVIDE BY FOUR

$$(\mathbb{E}[XY])^2 \leq \mathbb{E}[X^2] \mathbb{E}[Y^2]. \quad \blacksquare$$



With $\mathbb{E}[X^2] = 4$, $\mathbb{E}[Y^2] = 1$, $\mathbb{E}[XY] = 1.2$: the parabola floats above the axis

The best t . The lowest point is at $t = -\mathbb{E}[XY]/\mathbb{E}[Y^2]$, where $X + tY$ is the part of X that Y cannot explain. Its size, $\mathbb{E}[X^2] - \mathbb{E}[XY]^2/\mathbb{E}[Y^2]$, is the gap in disguise.

Cauchy–Schwarz in Disguise



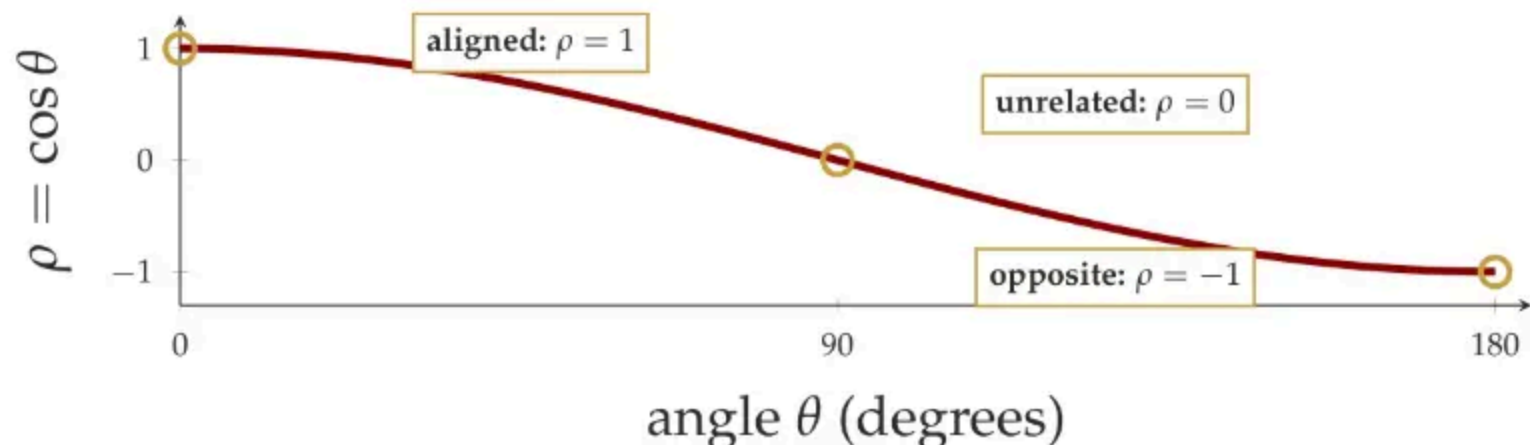
Choose X and Y cleverly and famous facts drop out. Nothing new is proved below: each row is the same inequality with different inputs.

Choice	It says	Known as
$Y = 1$	$(\mathbb{E}X)^2 \leq \mathbb{E}[X^2]$	$\text{Var}X \geq 0$
$ X , Y = 1$	$\mathbb{E} X \leq \sqrt{\mathbb{E}[X^2]}$	$L^1 \leq L^2$
centered X, Y	$\text{Cov}(X, Y)^2 \leq \text{Var}X \text{Var}Y$	$ \rho \leq 1$
finite lists	$(\sum a_i b_i)^2 \leq \sum a_i^2 \sum b_i^2$	sum form
functions	$(\int fg)^2 \leq \int f^2 \int g^2$	integral form
$X + Y$	$\ X + Y\ \leq \ X\ + \ Y\ $	triangle inequality

CORRELATION IS A COSINE

Centre both variables and define $\rho = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}$. Cauchy–Schwarz says exactly

$$\rho = \cos \theta \in [-1, 1].$$



No angle can push the correlation outside the band $[-1, 1]$

Triangle inequality for free. $\|X + Y\|^2 = \|X\|^2 + 2\mathbb{E}[XY] + \|Y\|^2 \leq (\|X\| + \|Y\|)^2$.

Measuring the Gap



How much room is left when the inequality is not tight? Take the list $a = (1, 2, 3)$, so $\sum a_i^2 = 14$, and test it against several lists b .

List b	$(\sum a_i b_i)^2$	$\sum a_i^2 \sum b_i^2$	Gap	$\cos \theta$
$(1, 2, 3)$	196	196	0	1
$(2, 4, 6)$	784	784	0	1
$(2, 1, 3)$	169	196	27	0.929
$(3, 0, -1)$	0	140	140	0
$(-1, -2, -3)$	196	196	0	-1

THE GAP IS AN ANGLE

Rearranging $\langle X, Y \rangle = \|X\| \|Y\| \cos \theta$ gives the exact identity

$$\|X\|^2 \|Y\|^2 - \langle X, Y \rangle^2 = \|X\|^2 \|Y\|^2 \sin^2 \theta.$$

THE GAP IS LEFTOVER SIZE

With $\beta = \mathbb{E}[XY]/\mathbb{E}[X^2]$, the part of Y that X cannot explain satisfies

$$\mathbb{E}[X^2] \mathbb{E}[(Y - \beta X)^2] = \mathbb{E}[X^2] \mathbb{E}[Y^2] - (\mathbb{E}[XY])^2.$$

So ρ^2 is the share of Y explained by X , and $1 - \rho^2$ is what is left.

Check. For $b = (2, 1, 3)$: $\rho^2 = 169/196 \approx 0.862$, so X explains 86.2% and 13.8% is left. The gap is $196 - 169 = 27$.

$$\text{gap} = \|X\|^2 \|Y\|^2 \sin^2 \theta$$

When the Rule Is Tight



A good inequality tells you when it becomes an equality. Here the gap closes exactly when the two vectors lie on one line.

THE EQUALITY CASE

Equality holds if and only if there are constants, not both zero, with

$$aX + bY = 0 \quad \text{almost surely.}$$

Then $Y = cX$, and $\rho = +1$ if $c > 0$ or $\rho = -1$ if $c < 0$.

Situation

What happens to the gap

$\theta = 0^\circ$ or 180°

gap is exactly 0

$\theta = 90^\circ$

gap is largest, $\|X\|^2\|Y\|^2$

Scale X by k

both sides grow by k^2 , ρ unchanged

Y is a multiple of X perfectly predictable

UNCERTAINTY PRINCIPLE

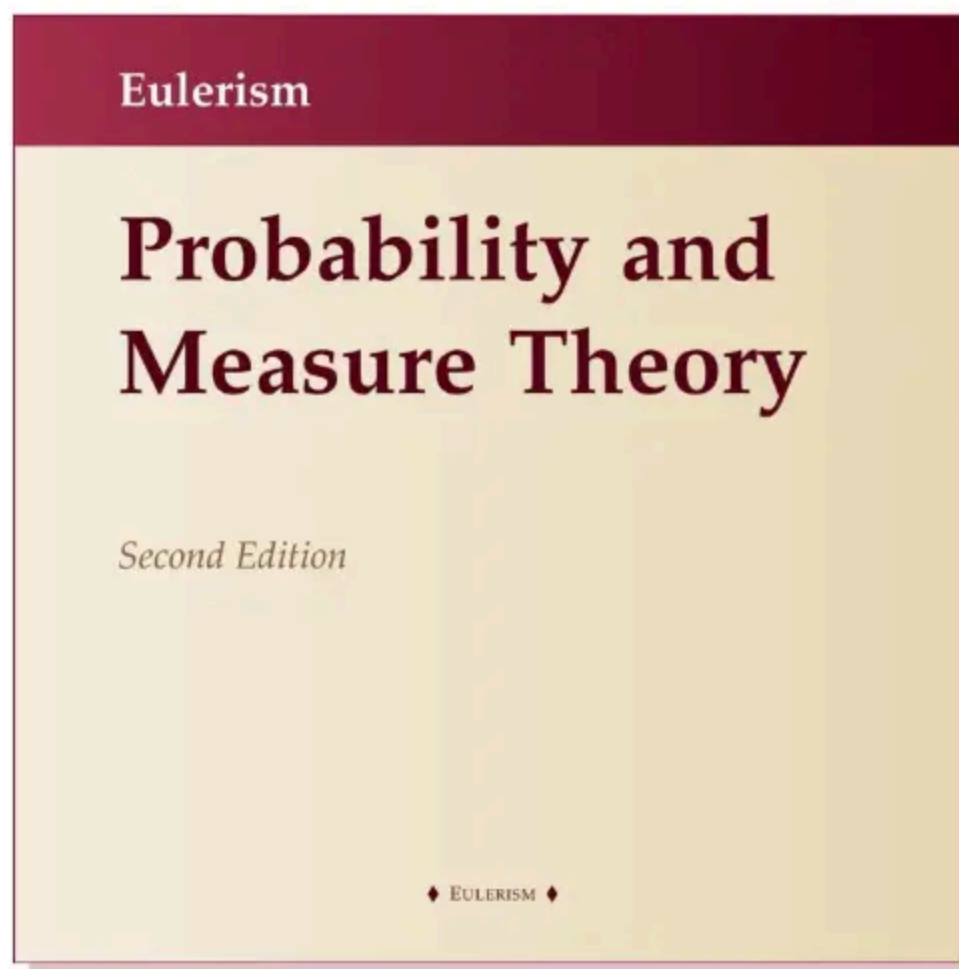
Apply Cauchy–Schwarz to the vectors $(A - \langle A \rangle)\psi$ and $(B - \langle B \rangle)\psi$ in a Hilbert space and you get

$$\Delta A \Delta B \geq \frac{1}{2} |\langle [A, B] \rangle|, \quad \Delta x \Delta p \geq \frac{\hbar}{2}.$$

Family tree. Hölder’s and Minkowski’s inequalities generalise it, and cosine similarity in machine learning is bounded by it. One cosine, a whole shelf of results.

Correlation is a cosine. It is just $(\mathbb{E}[XY])^2 \leq \mathbb{E}[X^2]\mathbb{E}[Y^2]$.

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EULERISM • PROBABILITY & MEASURE THEORY

The Most Beautiful Inequality In The Universe

$$\varphi(\mathbb{E}[X]) \leq \mathbb{E}[\varphi(X)]$$

*Bend first, average later, and you always
win more*

PRESENTED IN WORKS OF :



The Rule



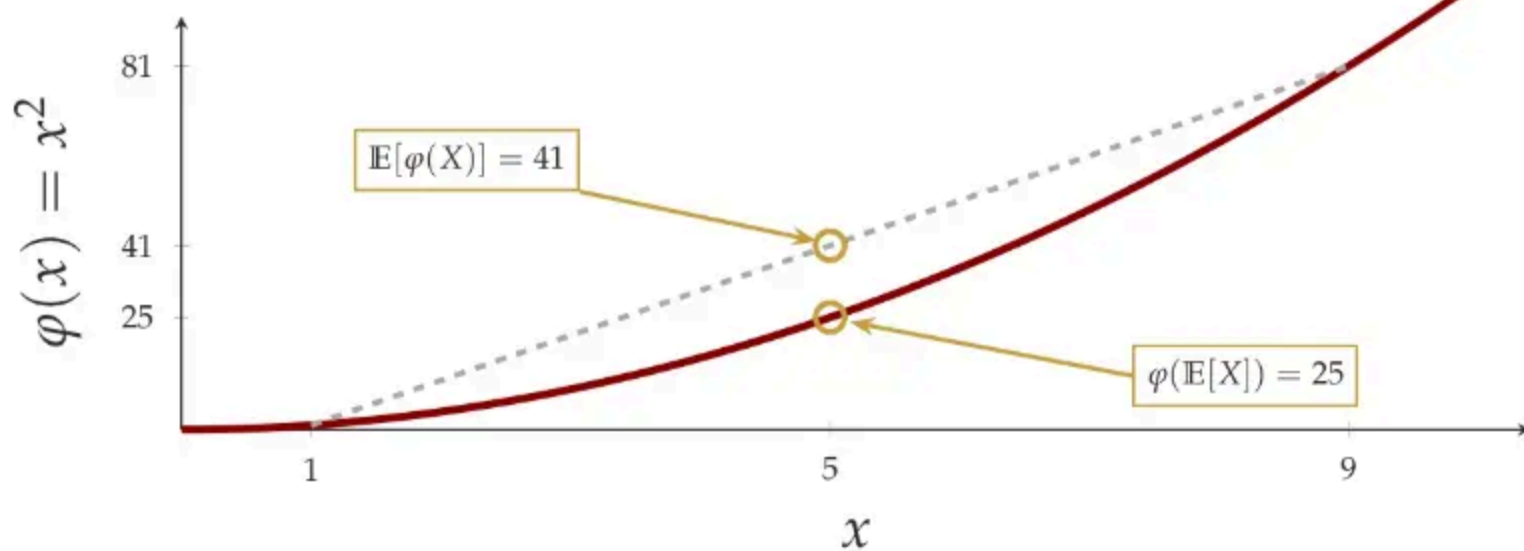
Take a random number X and squeeze it through a curve φ . You can average first and bend after, $\varphi(\mathbb{E}[X])$, or bend first and average after, $\mathbb{E}[\varphi(X)]$. For a convex curve the second order *always* wins. That is the whole rule.

JENSEN'S INEQUALITY

Let φ be convex, meaning the chord between any two points lies above the curve, and let X be a random variable with $\mathbb{E}|X| < \infty$. Then

$$\varphi(\mathbb{E}[X]) \leq \mathbb{E}[\varphi(X)].$$

For a concave φ the inequality flips.



$X \in \{1, 9\}$ with equal odds: the average sits on the curve, the chord sits above

$$\mathbb{E}[\varphi(X)] - \varphi(\mathbb{E}[X]) \geq 0 \quad (\varphi \text{ convex})$$

Why it is underrated. It looks like a trivial fact about curves, yet variance, the triangle inequality, AM–GM and the non-negativity of entropy are all Jensen in a different costume.

Why It Is True



The proof is three lines long and needs one picture: a convex curve never dips below its own tangent line.

STEP 1: TOUCH THE CURVE AT THE MEAN

Let $\mu = \mathbb{E}[X]$. Convexity gives a slope c (the derivative, when it exists) with

$$\varphi(x) \geq \varphi(\mu) + c(x - \mu) \quad \text{for every } x.$$

STEP 2: PLUG IN THE RANDOM VARIABLE

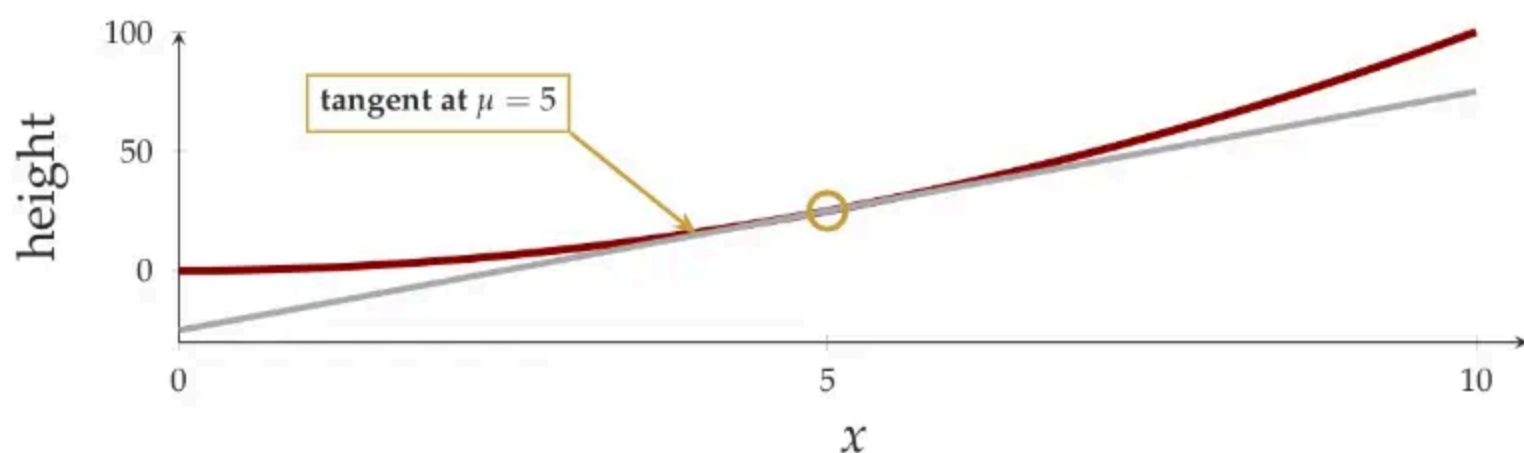
Replace x by X . The inequality holds outcome by outcome, so it survives taking expectations:

$$\mathbb{E}[\varphi(X)] \geq \varphi(\mu) + c(\mathbb{E}[X] - \mu).$$

STEP 3: THE LEFTOVER TERM DIES

Since $\mathbb{E}[X] - \mu = 0$,

$$\mathbb{E}[\varphi(X)] \geq \varphi(\mu) = \varphi(\mathbb{E}[X]). \quad \blacksquare$$



The curve never falls below its tangent, so averaging can only lift it

Linear things survive averaging. The tangent is a straight line, and $\mathbb{E}[a + bX] = a + b\mathbb{E}[X]$ exactly. All the slack in Jensen comes from the bending.

Jensen in Disguise



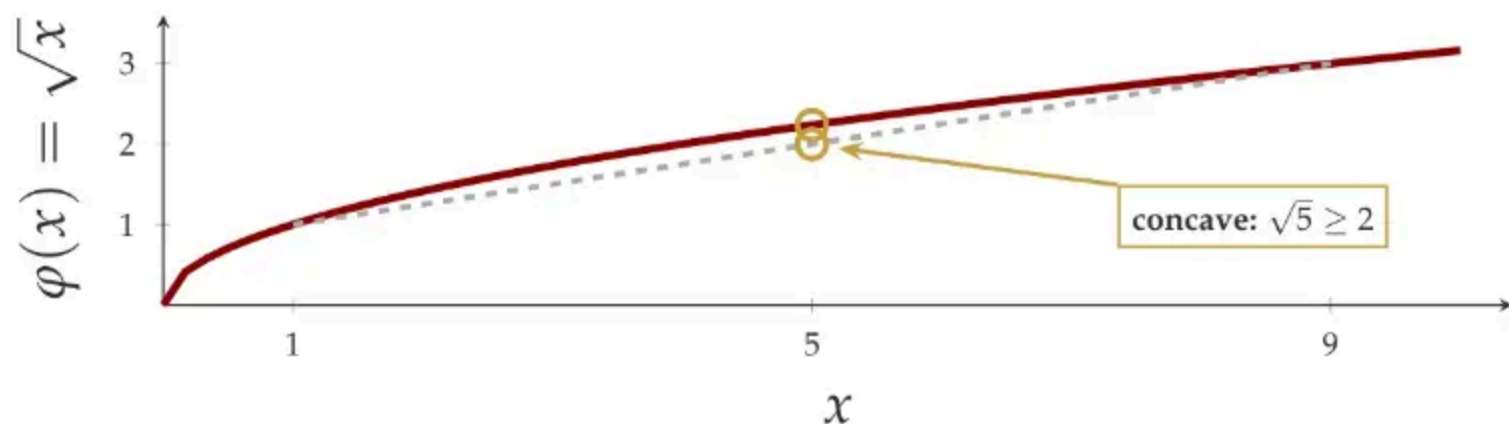
Choose a curve, and a famous inequality drops out. Nothing new is proved below: each row is Jensen with a different φ .

Curve $\varphi(x)$	Jensen says	Known as
x^2	$(\mathbb{E}X)^2 \leq \mathbb{E}[X^2]$	$\text{Var}X \geq 0$
$ x $	$ \mathbb{E}X \leq \mathbb{E} X $	triangle inequality
$-\ln x$	$\ln \mathbb{E}X \geq \mathbb{E} \ln X$	AM–GM
$1/x$	$1/\mathbb{E}X \leq \mathbb{E}[1/X]$	harmonic $\leq$ arithmetic mean
e^x	$e^{\mathbb{E}X} \leq \mathbb{E}[e^X]$	lower bound on the MGF
$x^p, p \geq 1$	$(\mathbb{E}X)^p \leq \mathbb{E}[X^p]$	power-mean inequality

VARIANCE IS A JENSEN GAP

Take $\varphi(x) = x^2$. The gap is not merely non-negative, it *is* the variance:

$$\mathbb{E}[X^2] - (\mathbb{E}[X])^2 = \text{Var}(X) \geq 0.$$



Bend the other way and the inequality flips: the chord now sits below

AM–GM in one line. With $X \in \{1, 9\}$: arithmetic mean 5, and $e^{\mathbb{E} \ln X} = \sqrt{1 \cdot 9} = 3$. Indeed $3 \leq 5$.

Measuring the Gap



Jensen only says the gap is non-negative. How large is it? Run one gamble, $X \in \{1, 9\}$ with probability $\frac{1}{2}$ each, so $\mathbb{E}[X] = 5$ and $\text{Var}X = 16$, through several curves.

Curve φ	$\varphi(\mathbb{E}X)$	$\mathbb{E}\varphi(X)$	Gap
x^2	25	41	+16
$1/x$	0.200	0.556	+0.356
$-\ln x$	-1.609	-1.099	+0.511
$\sqrt{x}$ (concave)	2.236	2.000	-0.236
$2x + 3$ (linear)	13	13	0

A RULE OF THUMB FOR THE GAP

Expand φ to second order around $\mu = \mathbb{E}[X]$ and take expectations:

$$\mathbb{E}[\varphi(X)] - \varphi(\mu) \approx \frac{1}{2} \varphi''(\mu) \text{Var}(X).$$

Two ingredients make the gap grow: more curvature, and more spread.

Check. For x^2 the formula gives $\frac{1}{2} \cdot 2 \cdot 16 = 16$, exact. For $-\ln x$ it gives $\frac{1}{2} \cdot \frac{1}{25} \cdot 16 = 0.32$, against the true 0.511: the estimate is rough when the spread is this wide.

WHY A RISK-AVERSE PERSON REFUSES THE COIN

Let wealth be 1 or 9 with equal odds, and let utility be the concave $u(w) = \sqrt{w}$. Jensen reverses: $\mathbb{E}[u(W)] = 2 < u(5) \approx 2.24$. The sure amount worth the same as the gamble solves $\sqrt{c} = 2$, so

$$c = 4 < \mathbb{E}[W] = 5.$$

The missing 1 is the price of risk, and it is a Jensen gap.

spread $\times$ curvature = the size of the gap

When the Rule Is Tight



A good inequality tells you when it becomes an equality. Here, the gap closes in exactly two situations.

THE EQUALITY CASE

If φ is *strictly* convex, then

$$\mathbb{E}[\varphi(X)] = \varphi(\mathbb{E}[X]) \iff X \text{ is almost surely constant.}$$

If φ is linear on the range of X , equality holds for *every* X .

Lever	What happens to the gap
Spread of X	grows like $\text{Var}X$
Curvature φ''	grows linearly with it
φ linear	gap is exactly 0
φ concave	sign flips, $\mathbb{E}\varphi(X) \leq \varphi(\mathbb{E}X)$
X constant	gap is exactly 0

ENTROPY IS NEVER NEGATIVE

Let P and Q be distributions with Q positive wherever P is. Apply Jensen to the convex $-\ln$ with $X = Q/P$ under P :

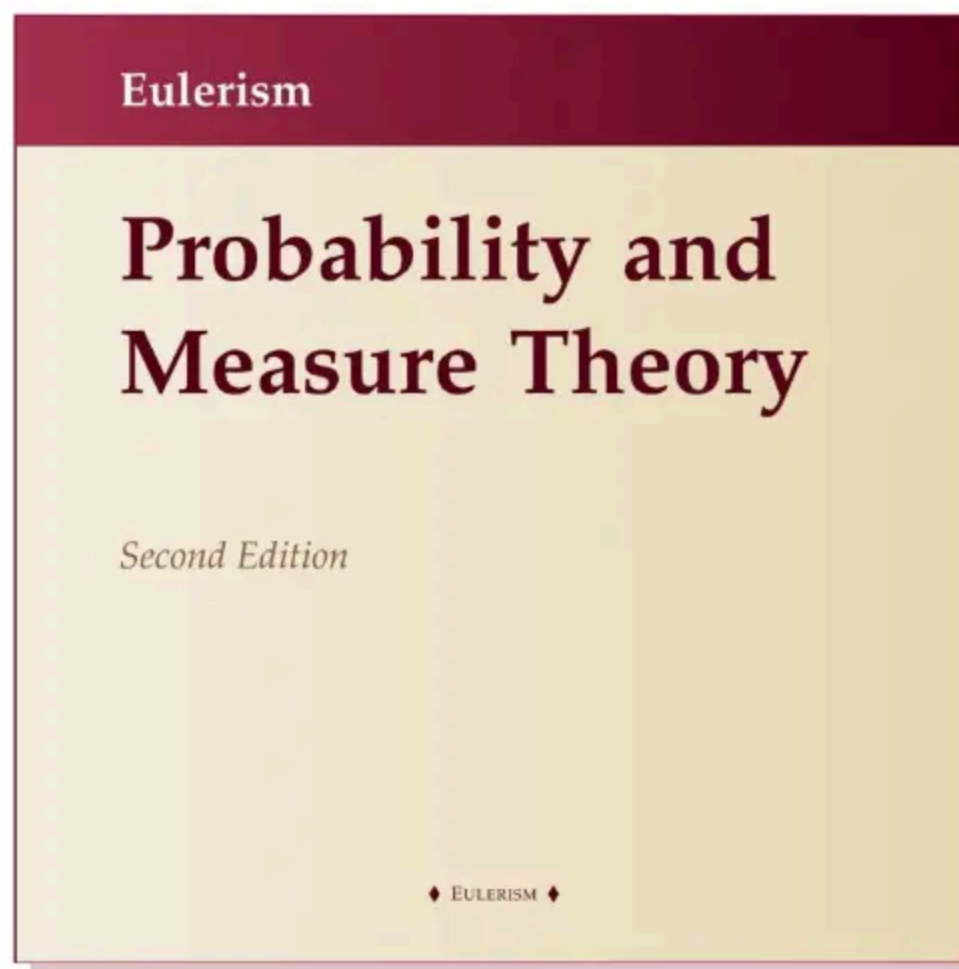
$$D(P \parallel Q) = \mathbb{E}_P\left[-\ln \frac{Q}{P}\right] \geq -\ln \mathbb{E}_P\left[\frac{Q}{P}\right] = -\ln 1 = 0.$$

Relative entropy is non-negative because a logarithm bends.

Family tree. Hölder's inequality, Lyapunov's inequality and Gibbs' inequality all descend from the same convexity. Learn this one rule well and a whole shelf of inequalities becomes a single idea.

Averaging hides the spread. It is just $\varphi(\mathbb{E}[X]) \leq \mathbb{E}[\varphi(X)]$.

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